

Form B2 Software Disclosure Statement

UL ref: _____

(applies to computer programs; for patentable inventions, please use Form B1)

Submit to:	☐	Technology Transfer & Fundraising	ktt@uni.lu
	☐	Technology Transfer / LCSB	lcsb-tto@uni.lu
	☒	Technology Transfer / SnT	snt-tto@uni.lu

For TTO use only:

Form A received on:		Form B received on:	
Research unit:		Laboratory:	
Case Manager:		Deputy Case Manager:	
Receipt acknowledged to each author			
Original signatures of ALL authors, received on (date):			
U.S. assignment from each author to University, received on (date):			
Electronic copy of ENTIRE software received (location and date)			
Public funding	NO ☐	YES ☐	Grant No.: _____
External Collaborator:			IPR agreement: _____
New Inter-Institution Agreement (IIA) required?			NO ☐ YES ☐

To be filled in by inventors / research unit:

Working title of the invention, or acronym:

AGRI-Fender

Research Unit / Lab	SEDAN		
Time period during which the inventor(s) conceived the invention:			January 2024 – April 2024
Head of RU informed	Name:	Radu State	Date: 15/05/2024
			Signed: _____

PART 1 – AUTHORS

(Please remember to fill in the complete details for each author in Part 6 at the end of this form.)

N°	Author (name)	Institution (name)	Contribution* (%)
1	Hui Huang	SnT	25%
2	Miguel Kiyoshy Nakamura Pinto	SnT	25%
3	Jeovane Honorio Alves	SnT	25%
4	Radu State	SnT	25%

** Contribution to the computer programme. This information is provided by the researchers/authors involved in the project; therefore the University of Luxembourg assumes no responsibility for the accuracy of this declaration.*

PART 2 – SOFTWARE INFORMATION

2.1 Technical Field

As an introduction, please describe briefly the (technical) field to which your software belongs:

Our solution addresses a robot that can detect landmines using radar technology and image processing.

2.2 Technical Summary of the Software

Please describe here the software as precisely as possible: (a) motivation for software, problem solved, opportunity addressed, (b) usefulness, (c) advantages and disadvantages when comparing to alternatives, (d) best use and variations, (e) drawings, schematics and flowcharts, if appropriate.

Motivation for Software:

Ukraine is currently experiencing a situation where agricultural lands are being mined in ongoing conflicts, requiring immediate and ongoing demining interventions. Ukraine is one of the world's largest grain producers, and the presence of mines severely disrupts this production, with global consequences for food prices and supply. Agricultural demining in Ukraine is time-sensitive due to the country's seasonal climate; land must be cleared in time for planting and harvesting cycles. Demining is also critical for the protection of agricultural infrastructure, such as storage facilities and transport routes, which are essential for the distribution of produce. Given the current and pervasive threat in Ukraine, there is a pressing need for innovative demining technologies that can be rapidly deployed and operated in the context of an active conflict zone.

Problem Solved:

The proposed technical solution is an integrated system that enhances the capability of small robotic platforms for the purpose of humanitarian demining, specifically focusing on the detection and mapping of antipersonnel mines. The system combines advanced sensing technologies with communication and collaboration tools to create a comprehensive demining solution. It comprises the following components:

- **Ground Penetrating Radar (GPR):** A miniaturized GPR system mounted on the robot utilizes high-frequency radar pulses to detect the presence of mines underground. The radar's sensitivity allows for the identification of objects with the density and size consistent with antipersonnel mines. This is implemented using a Software Defined Radio (SDR): The SDR component digitally processes the radar signals, providing a flexible and upgradable platform for the radar operation. The software can be updated to adapt to different soil conditions and to distinguish between mine signatures and other metallic objects. Implementation and code was developed at SnT.
- **360 Panoramic Camera:** The robot is equipped with a high-resolution camera capable of taking panoramic images for surface surveillance. This helps in detecting surface-laid mines, tripwires, and provides visual confirmation of the terrain. The camera is trained on images of landmines and uses SnT developed software to recognise in 360 images landmines, but training of the model was done on available 2D images. This is a unique value proposition in our proposal.
- **Mapping and Threat Identification Software:** The onboard system processes data from the GPR and the camera to create real-time maps of the area. It tags detected threats with GPS coordinates, creating an annotated map that marks safe zones and areas where mines are likely present.
- **Collaborative Demining Workflow:** The robot is integrated into a network that allows for real-time communication with other demining units and control centers. Deminers can control multiple robots simultaneously and share maps and data to coordinate demining efforts efficiently. This can be done using a cloud based solution or alternatively using a server on the demining robot.
- **User Interface (UI):** A user-friendly interface allows deminers to monitor the progress of each robot, review the mapped data, and plan safe demining operations. The UI can display multiple data layers, including real-time video feeds, GPR data, and annotated threat maps.
- **Low cost robotic/hover frames,** which allow to swarm a demining area with expandables robots.

- Active learning: Agri-Fender allows users to annotate data during operations and allow to update the training model.

Opportunity Addressed:

The market for a humanitarian demining robot is diverse and spans multiple sectors, each with a vested interest in the safe and effective removal of landmines and unexploded ordnance (UXO). The primary beneficiaries and end-users for such technology can be categorized as follows:

- Public authorities, certified demining operators and international aid groups engaged in demining activities are primary users. These organizations often operate in post-conflict zones where they strive to restore safety and facilitate the return of displaced populations.
- Public and local authorities responsible for public safety, land management, and development projects would use such robots to clear lands and prepare them for civilian use or development projects.
- Farmers and agricultural cooperatives in post-conflict areas would benefit from the use of demining clearance these robots to reclaim arable land, which is vital for food production and economic sustainability.
- Companies involved in the development of infrastructure, housing, and commercial projects in areas would use this technology the demining clearance to ensure safe construction sites

Usefulness:

The solution implements the following workflow:

Robots are deployed into a suspected minefield, where they navigate autonomously or are remotely guided by operators. As the robots traverse the area, the GPR and cameras continuously scan the ground and surroundings. The GPR identifies potential mines below the surface, while the camera spots surface-level hazards. Signals from the GPR are processed in real-time through the SDR to classify objects and differentiate between mine signatures and false positives. The panoramic images are analyzed using computer vision algorithms to detect visual indicators of mines. Detected threats are tagged on the map with their GPS coordinates. The map is updated in real-time and can be accessed by operators through the UI. Deminers can work collaboratively by sharing the maps and sensor data. They can assign different robots to survey specific areas, cross-reference findings, and validate detections. Using the annotated maps, deminers plan the safest routes for manual verification and neutralization of mines. After an area has been cleared, the system generates a comprehensive report detailing the operation, findings, and status of the area.

Advantages When Comparing to Alternatives:

- Innovative Technology: Agri-Fender allows to fuse both visual and radar data and allow active updating of training data
- Cost-Effectiveness: Agri-Fender is more affordable and thus key to adoption by NGOs and developing nations.
- Adaptability and Scalability: Agri-Fender uses off the shelves hardware, which allows for short supply chains and easily replaceable hardware.
- Collaborative Operations: There is growing interest in systems that can work in fleets or in conjunction with humans and other machines, enhancing efficiency and coverage. Agri-Fender addresses this need.

Disadvantages When Comparing to Alternatives:

- Not yet tested with real landmines.

Best Use and Variations:

While the primary focus is humanitarian, military entities could utilize this technology for demining in conflict or post-conflict zones to protect personnel and ensure safe military operations.

Entities like the United Nations, World Bank, and various development funds could deploy these robots as part of broader initiatives to support sustainable development in post-conflict nations.

International peacekeeping troops often operate in areas where they face the threat of landmines and UXO. Such robots could be used to secure areas and ensure the safety of both the peacekeepers and the local population.

By providing a safer, more efficient, and potentially more cost-effective means of demining, such robots address a critical global issue. The technology holds the promise of not only saving lives but also transforming land from hazardous to habitable, enabling economic development and the rebuilding of communities long after conflicts have ended.

2.3 Third Party Rights

Does the software include any of the following code which is:

(Please join to this form any relevant documents.)

...protected by a patent	NO
...downloaded from the web (indicate pre-existing licences for each code or package you have used)	
Code/package	FOSS licence
See SWP-dependencies_package.json (attached)	All MIT license
...brought to the UL from a company or a different university	NO
...developed by a collaborator at a different institution	NO
...obtained under a licence	NO
...unable to be claimed under authorship or ownership	NO
...an enhancement or derivative works of previously developed software	NO

2.4 Distribution Request

Are you requesting distribution of this software under a specific licence or distribution?

No

2.5 Future Development

Do you intend to make enhancements to the software?

Yes

PART 3 – SOFTWARE CREATION FRAMEWORK

3.1 Financial Support

What is the funding support of the work that led to the software?

Fully supported by UL	FNR	FP 7	Other E.U.	ESA	Contract with a company	Contract with public authorities	Other partner
THREE PEOPLE WERE FULLY FUNDED BY UL	ONE PERSON WAS PARTIALLY FUNDED BY FNR						

FNR / EU / ESA (or other) grant No.: *(must be cited in any publication)*

Not Applicable

3.2 External Collaborations

Please provide details and the role of all external contact (Universities, private companies, etc.) related to your software

Not Applicable

Has an NDA or a MTA been signed with these partners ?

☐ Yes x No

Has another contract been signed ?

☐ Yes x No

If Yes, please provide a copy of these agreements.

PART 4 – PRIOR PUBLIC DISCLOSURE¹

Have you already publicly disclosed your software outside of the University of Luxembourg?

Where	X	Details (where, when, to whom, how)
In a publication		
In a poster or summary		
In an oral presentation		
In the Internet (forum, ...)		
During a meeting		
During a PhD defence		
During a seminar		
During a conference		
Other (please specify)		

Please join to this form a copy of all documents (conference or seminar proceedings, meeting reports, print-outs, etc.) linked to the disclosure.

¹ Public disclosure includes any article, any written notice (such as poster for a conference), any oral communication (presentation in a conference, discussion with a business partner, etc.), wherever in the world, to third persons without any protection agreements (NDA or MTA signed)

(PLEASE COMPLETE THIS SECTION 5 ONLY IF THE SOFTWARE HAS COMMERCIAL POTENTIAL AND/OR COULD POSSIBLY GENERATE REVENUE FROM LICENSING)

PART 5 – COMMERCIAL POTENTIAL

5.1 Application

What are the potential application fields of for your software?

Fields	X	Details
Life science or biotechnology		
Material science or engineering		
ICT		
Manufacturing	X	This technology can be used in machine-building industry
Automotive, aircraft or space		
Legal, finance or economy		
Education, training or human resources		
Public policy or society	X	Sells to public authorities
Other (please specify)		

Please join to this report a copy of all relevant documents.

5.2 How were the needs addressed by the software satisfied until now?

Several companies and research institutions have developed technologies to improve the efficiency and safety of demining operations. Some notable competitors include:

1. Mine Kafon Drone: An aerial drone that detects and detonates landmines using a robotic arm. This technology emphasizes the use of drones to minimize human risk.
2. Clearpath Robotics: Known for its unmanned vehicle platforms, Clearpath offers customizable robots which can be equipped with various sensors and tools for demining.
3. DOK-ING MV-4: A Croatian company that produces the MV-4 Mine Clearance System, a heavy-duty, remote-controlled vehicle designed for clearing all types of mines and UXO.
4. Hydronalix : Offers robotic solutions that use sonar and other technologies for underwater mine detection, showing the diversity in operational environments.

Different competitors employ varying technologies, each with its unique advantages and limitations:

- Aerial vs. Ground-Based Systems: Aerial systems like the Mine Kafon Drone can quickly cover large areas but may struggle with accuracy and the ability to handle larger, buried UXO. Ground-based systems like those from Clearpath and DOK-ING offer thorough scanning and the ability to deal with rugged terrain.
- Autonomy Levels: The degree of autonomy varies significantly among competitors, affecting the need for human oversight and the associated risks.
- Sensor Technology: Technologies range from basic metal detectors to advanced GPR and infrared sensors, affecting the ability to detect non-metallic mines.

5.3 What could be the competitive advantages of your software?

Does the software enhance the quality or value of an existing product or service? Does the software allow an existing product or service to be offered in a more economical manner? What will it cost to bring the software to the market? What price can the software be sold at? What sales can be expected?

- Innovative Technology: AGRI-Fender allows to fuse both visual and radar data and allow active updating of training data
- Cost-Effectiveness: AGRI-Fender is more affordable and thus key to adoption by NGOs and developing nations.
- Adaptability and Scalability: AGRI-Fender uses off the shelves hardware, which allows for short supply chains and easily replaceable hardware.
- Collaborative Operations: There is growing interest in systems that can work in fleets or in conjunction with humans and other machines, enhancing efficiency and coverage. Agri-Fender addresses this need.

5.4 What could be the limitation(s) of your software compared with existing products?

We have not yet tested it on real landmines yet.

5.5 Are you aware of any obstacles to commercial exploitation?

(technical problem, problem with the IPR, cost of implementation for your partner, normative obstacle, legal issues, resistance to change, ...)

No

5.5 Please describe a potential market:

Humanitarian demining is a critical need in various parts of the world where conflicts have left a legacy of landmines and unexploded ordnance (UXO). Focusing specifically on Africa and Ukraine, both regions have unique challenges and needs in relation to agriculture, which is a key factor in their socio-economic development and food security. Several African countries are affected by landmines and UXO from past conflicts, which continue to pose a significant threat to rural communities, often the most dependent on agriculture for their livelihoods. Vast areas of arable land across Africa are unusable due to the presence of landmines. Demining efforts are essential for the reclamation of this land for agricultural purposes, to combat food insecurity and to promote self-sufficiency. Clearing landmines would enable not just farming activities but also the construction of infrastructure such as roads and irrigation systems, which are vital for agricultural development and market access. Mines also threaten protected areas and wildlife. Their removal is crucial to safeguard the biodiversity which is often integral to the ecological balance that supports agriculture.

PART 6 - AUTHORS**AUTHOR (1)**

Contribution to the software: 25 %

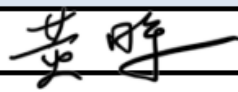
Staff Nr				
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First name	Hui			
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Faculty & Research unit	
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Place	Date	Signature
Luxembourg	22/05/2024	

AUTHOR (2)

Contribution to the software:

25

%

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Place	Date	Signature
Luxembourg	22/05/2024	

AUTHOR (3)

Contribution to the software:

25

%

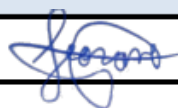
Staff Nr				
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Place	Date	Signature
Luxembourg	22/05/2024	

AUTHOR (4)

Contribution to the software:

25

%

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University / Organisation	
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Place	Date	Signature
Luxembourg	22/05/2024	

Dependencies of the software:

```

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